

Surface-Object Detection and Tracking from Commercial X-Band Marine Radar Using Adaptive ST-DBSCAN

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Maritime missions such as over-water navigation, ship-based operations, and airborne/space-enabled maritime surveillance require reliable surface-object detection and tracking in the presence of strong, nonstationary sea clutter and intermittent object returns. We present an end-to-end, real-time pipeline for vessel detection and multi-sweep tracking using commercial X-band radar providing only intensity measurements (range, azimuth, and intensity) without Doppler phase history. Raw polar returns from a Furuno NXT solid-state unit are transformed into Cartesian point clouds and clustered using ST-DBSCAN to link detections across sweeps. Neighborhood radii adapt automatically via k -nearest-neighbor distance statistics to handle varying clutter and point densities. A lightweight gap-recovery mechanism reconnects clusters across brief temporal dropouts. We evaluate the approach on (i) a synthetic dataset with ground-truth trajectories and (ii) coastal radar data from a shore-mounted installation, characterizing detection-to-track association behavior relative to baseline DBSCAN and standard ST-DBSCAN under nonstationary clutter and intermittent returns. A Rust implementation runs at 2 sweeps/s on an NVIDIA Jetson Orin, exceeding real-time requirements for a 48 RPM radar.

I. Nomenclature

ε_s	=	spatial neighborhood radius for clustering (pixels)
ε_t	=	temporal neighborhood radius (number of frames)
$N_{\min}$	=	minimum number of neighbors required for a core point
k	=	number of nearest neighbors for adaptive radius estimation
$d_k(\mathbf{p})$	=	distance to k -th nearest neighbor of point $\mathbf{p}$
$d(\mathbf{p}, \mathbf{q})$	=	Euclidean distance between points $\mathbf{p}$ and $\mathbf{q}$
$\mathbf{p}_i$	=	radar point with coordinates (x_i, y_i) and frame index t_i
$d_{\max}$	=	maximum distance threshold for track-to-cluster association
d_{gap}	=	maximum centroid distance for gap recovery (pixels)
$\tau_{\max}$	=	maximum track age before deletion (frames)
τ_{gap}	=	maximum gap duration for cluster reconnection (frames)

II. Introduction

When a small vessel enters a wave trough off the South Carolina coast, its radar return vanishes for two or three antenna sweeps. A conventional tracking system, seeing no detections, increments the track's age counter; after several more dropouts caused by wave shadowing and radar cross-section fluctuations, the system deletes the track entirely. Moments later, the vessel reappears—but now as a new track with a fresh identifier, its previous history lost. This scenario, repeated across dozens of vessels during a typical harbor approach, degrades situational awareness precisely when it matters most.

Maritime domain awareness is essential for navigation safety, port security, and search-and-rescue operations. Commercial X-band marine radars operating at 9.3–9.5 GHz provide reliable all-weather sensing of surface vessels and other objects of interest. However, extracting consistent object tracks from raw radar data remains challenging due to sea clutter, multipath reflections, and the intermittent target visibility illustrated above.

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Traditional radar processing pipelines employ Constant False Alarm Rate (CFAR) detection followed by kinematic filtering using Kalman or alpha-beta trackers [4, 5]. While effective for isolated targets in open water, these methods struggle with dense vessel traffic and the nonstationary clutter conditions common in coastal waters. CFAR detectors require careful threshold tuning for each environment, and single-frame detection discards valuable temporal information that could link fragmented returns across sweeps.

The increasing availability of solid-state marine radars with high range resolution (sub-meter) and fast scan rates (24–48 RPM) motivates point-cloud-based processing approaches that exploit both spatial and temporal coherence. Unlike traditional plot extraction, which reduces each target to a single position estimate per scan, point cloud methods preserve the spatial extent of radar returns, enabling better discrimination of closely spaced targets and more robust association across frames.

This paper presents an end-to-end detection and tracking system based on an adaptive variant of Spatio-Temporal DBSCAN (ST-DBSCAN) clustering. The main contributions include an adaptive ST-DBSCAN formulation where neighborhood radii are estimated automatically from k-nearest-neighbor distance statistics, eliminating manual parameter tuning for different clutter environments; a gap-recovery mechanism that reconnects clusters across brief temporal dropouts caused by intermittent radar returns; an efficient implementation combining spatial hashing for $O(1)$ average-case neighbor queries with union-find for cluster merging; and experimental evaluation on synthetic scenarios with ground-truth trajectories and real coastal radar data, characterizing tracking behavior under varying conditions.

III. Related Work

A. Radar Target Detection

Traditional marine radar processing uses CFAR detection to identify targets against sea clutter and thermal noise [4]. Cell-averaging CFAR (CA-CFAR) estimates local noise power from neighboring range cells and sets a detection threshold accordingly. While effective in homogeneous clutter, CA-CFAR suffers from target masking in dense traffic and requires environment-specific tuning.

Recent advances in solid-state radar technology have motivated point-cloud-based processing approaches. Vivone et al. [6] demonstrated multi-target tracking from synthetic aperture radar using density-based clustering. In the automotive domain, radar point cloud processing has benefited from deep learning approaches, though these require extensive labeled training data not typically available for maritime applications.

B. Density-Based Clustering

DBSCAN (Density-Based Spatial Clustering of Applications with Noise), introduced by Ester et al. [1], identifies clusters as regions of high point density separated by regions of low density. The algorithm classifies each point as either a *core point* (having at least $N_{\min}$ neighbors within distance ϵ), a *border point* (within ϵ of a core point but not itself a core point), or *noise*.

ST-DBSCAN [2] extends DBSCAN to spatio-temporal data by introducing separate thresholds for spatial (ϵ_s) and temporal (ϵ_t) dimensions. Points must satisfy both spatial and temporal proximity to be considered neighbors, preventing merging of spatially close but temporally distant points—essential for tracking moving objects.

A key limitation of standard DBSCAN is sensitivity to the choice of ϵ . Several extensions address this: OPTICS [7] produces a reachability ordering that can be cut at various density thresholds, while HDBSCAN [8] constructs a cluster hierarchy and extracts stable clusters automatically. Our k-nearest-neighbor approach to adaptive ϵ estimation is related to the common “elbow method” for DBSCAN parameter selection, but we apply it in real-time to each frame rather than as an offline preprocessing step.

C. Multi-Object Tracking

Multi-object tracking (MOT) maintains object identities across time by solving the data association problem—matching new detections to existing tracks [5]. The Hungarian algorithm [3] provides a globally optimal solution in $O(n^3)$ time.

Joint Probabilistic Data Association (JPDA) [9] handles measurement-to-track association uncertainty by computing weighted updates across all plausible associations. For extended targets producing multiple measurements, Probability Hypothesis Density (PHD) filters [10] provide principled approaches to multi-object tracking. Recent work on Simple Online and Realtime Tracking (SORT) [11] demonstrates that combining simple motion models with optimal assignment

can achieve competitive performance at real-time rates, motivating our Hungarian-based approach.

IV. Methodology

The processing pipeline consists of four stages: (1) point cloud extraction from radar imagery, (2) adaptive parameter estimation, (3) spatio-temporal clustering with gap recovery, and (4) multi-object tracking. Figure 1 illustrates the system architecture.

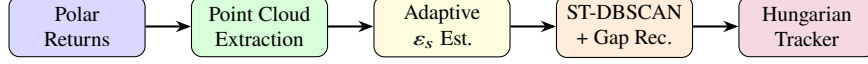


Fig. 1 Processing pipeline from polar radar returns to tracked objects.

A. Point Cloud Generation

The Furuno DRS-NXT radar outputs Plan Position Indicator (PPI) images as 8-bit grayscale bitmaps representing received signal intensity. Unlike Doppler radars, the unit provides only intensity measurements without phase history. The preprocessing stage converts polar radar data to Cartesian point clouds through four operations. First, raw returns in (r, θ) coordinates are converted to Cartesian (x, y) using standard trigonometric projection. Second, intensity thresholding discards pixels with intensity $I < I_{\text{thresh}}$ where $I_{\text{thresh}} = 25$ (10% of maximum for 8-bit data). Third, stride-based subsampling with stride $s = 2$ reduces point count by 75% with negligible impact on clustering quality. Finally, each point is assigned a frame identifier t_i corresponding to the radar sweep number.

Figure 2 illustrates the point cloud extraction process. A typical sweep produces 12,000–50,000 points after preprocessing.

B. Adaptive Parameter Estimation

Standard ST-DBSCAN requires manual selection of ε_s , which is sensitive to local point density. We propose automatic estimation using k-nearest-neighbor (k-NN) distance statistics.

For each point $\mathbf{p}$, let $d_k(\mathbf{p})$ denote the distance to its k -th nearest spatial neighbor. The distribution of d_k values across all points characterizes local density. We estimate ε_s as:

$$\varepsilon_s = \text{percentile}_{90}(\{d_k(\mathbf{p}) : \mathbf{p} \in \mathcal{P}\}) \quad (1)$$

where $k = N_{\min}$ (typically 5). This choice ensures that approximately 90% of points have sufficient neighbors to potentially become core points, while rejecting the sparsest 10% as likely noise.

The k-NN distances are computed efficiently using the same spatial hash structure used for clustering, yielding $O(n)$ average-case complexity for parameter estimation.

C. ST-DBSCAN Clustering

The neighborhood of a point $\mathbf{p}$ is defined as:

$$\mathcal{N}(\mathbf{p}) = \{\mathbf{q} \in \mathcal{P} : d(\mathbf{p}, \mathbf{q}) \leq \varepsilon_s \wedge |t_{\mathbf{p}} - t_{\mathbf{q}}| \leq \varepsilon_t\} \quad (2)$$

A point is a *core point* if $|\mathcal{N}(\mathbf{p})| \geq N_{\min}$. Clusters form by connecting density-reachable core points. Border points join their nearest core point's cluster; isolated points are labeled as noise.

Setting $\varepsilon_t = 2$ links detections across five consecutive frames, improving cluster coherence when individual frames have sparse returns.

1. Spatial Hashing

We use a spatial hash grid that partitions the 3D space (x, y, t) into cells of size ε_s :

$$\text{cell}(\mathbf{p}) = \left(\left\lfloor \frac{x}{\varepsilon_s} \right\rfloor, \left\lfloor \frac{y}{\varepsilon_s} \right\rfloor, t \right) \quad (3)$$

All spatial neighbors lie within 27 cells (3×3×3 in space-time). For uniformly distributed points, this yields $O(1)$ average-case neighbor queries.

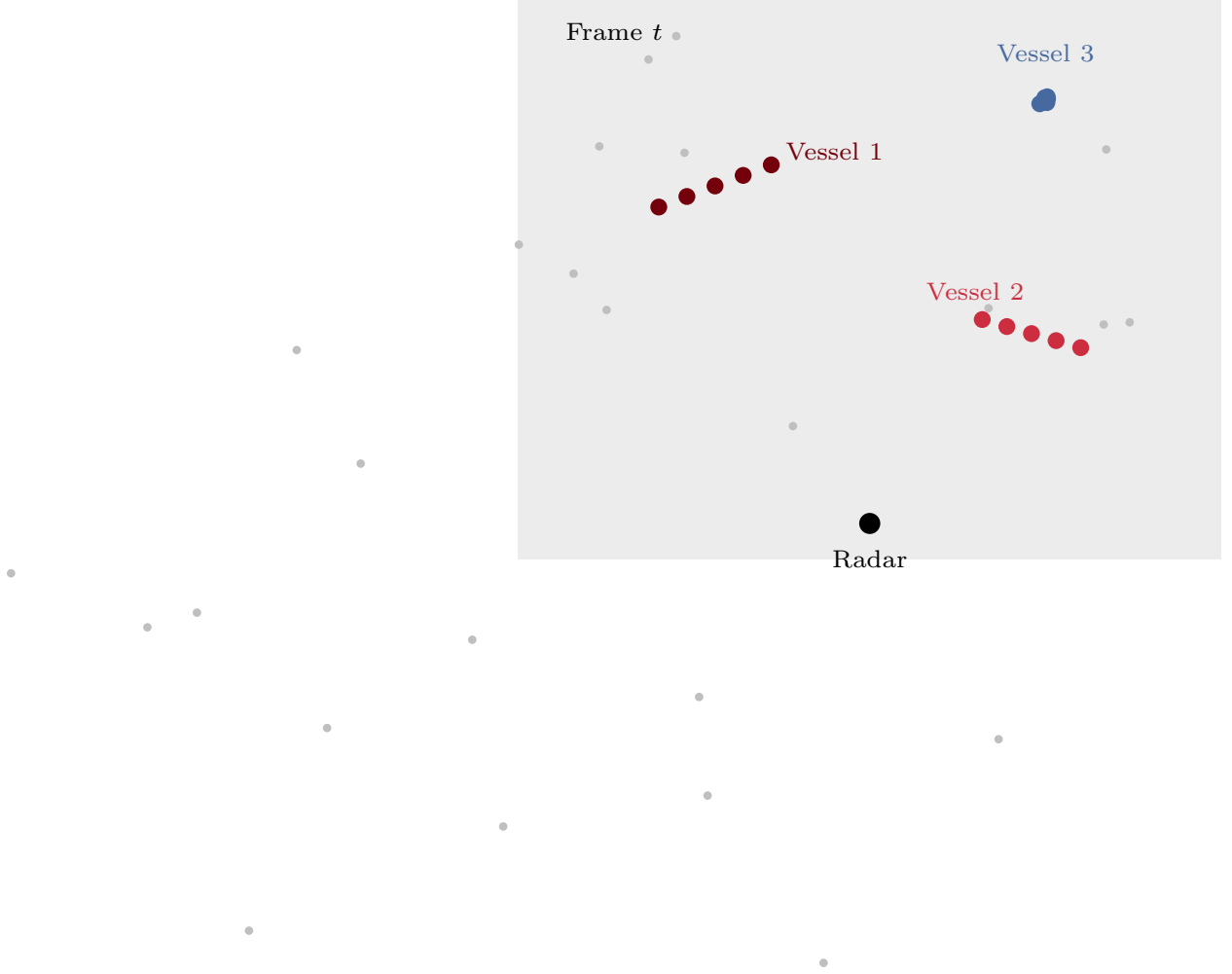


Fig. 2 Point cloud extraction: (a) raw radar PPI, (b) binary mask after intensity thresholding, (c) resulting point cloud with intensity-coded colors.

2. Union-Find for Cluster Merging

Connected core points merge using a union-find data structure with path compression and union by rank, yielding $O(\alpha(n))$ amortized time per operation where α is the inverse Ackermann function.

D. Gap Recovery Mechanism

Intermittent radar returns cause temporal gaps where a vessel produces no detectable points for one or more sweeps. Standard ST-DBSCAN with $\varepsilon_t = 2$ handles gaps up to 2 frames, but longer dropouts fragment clusters. We propose a lightweight gap-recovery post-processing step in which we first identify cluster pairs (C_i, C_j) where C_i ends at frame t_1 and C_j begins at frame t_2 with temporal separation $t_2 - t_1 \leq \tau_{\text{gap}}$ (gap threshold, typically 5 frames) and spatial centroids satisfying $\|\bar{\mathbf{x}}_i - \bar{\mathbf{x}}_j\| < d_{\text{gap}}$. Qualifying cluster pairs are then merged by relabeling C_j points with C_i 's cluster ID. This reconnects clusters across dropouts of up to τ_{gap} frames without modifying the core ST-DBSCAN algorithm.

E. Multi-Object Tracking

The tracking module maintains persistent object identities using a predict-associate-update cycle.

1. Motion Model

Each track maintains a velocity estimate using exponential smoothing:

$$\mathbf{v}_k = \alpha \cdot (\mathbf{x}_k - \mathbf{x}_{k-1}) + (1 - \alpha) \cdot \mathbf{v}_{k-1} \quad (4)$$

where $\alpha = 0.7$. The predicted position is $\hat{\mathbf{x}}_{k+1} = \mathbf{x}_k + \mathbf{v}_k$.

2. Hungarian Assignment

The cost of assigning track i to cluster j is the Euclidean distance $C_{ij} = \|\hat{\mathbf{x}}_i - \mathbf{c}_j\|$. The Hungarian algorithm finds globally optimal one-to-one assignments in $O(m^3)$ time.

3. Track Lifecycle

Track management follows a three-state lifecycle: unmatched clusters spawn new tracks upon first detection; matched tracks update their position and velocity estimates while resetting their age counter; and tracks exceeding $\tau_{\max}$ consecutive unmatched frames are deleted from the active track set.

V. Implementation

The system is implemented in Rust (approximately 6,000 lines) for performance and memory safety. Table 2 lists the default configuration parameters.

Table 2 Default Configuration Parameters

Parameter	Symbol	Default Value
k-NN for adaptive ε_s	k	5
Temporal radius	ε_t	2 frames
Min neighbors	$N_{\min}$	5 points
Gap threshold	τ_{gap}	5 frames
Gap recovery dist.	d_{gap}	50 pixels
Max association dist.	$d_{\max}$	50 pixels
Max track age	$\tau_{\max}$	5 frames
Velocity smoothing	α	0.7

The implementation uses the Rayon library for parallel neighbor discovery and an optional WebGPU (wgpu) backend for GPU-accelerated distance computation.

VI. Experimental Results

We evaluate the approach on synthetic data with ground-truth trajectories and real coastal radar recordings.

A. Synthetic Dataset

We created three synthetic scenarios of increasing difficulty to systematically characterize tracking behavior:

Scenario A (Easy): 3 vessels on parallel linear trajectories, well-separated (>100 px), constant velocity (5 px/frame), 100 sweeps. Gaussian position noise $\sigma = 2$ px, 50 clutter points per frame.

Scenario B (Medium): 5 vessels including 2 crossing paths and one 90° turn, velocities 3–8 px/frame, 150 sweeps. Position noise $\sigma = 3$ px, 100 clutter points per frame.

Scenario C (Hard): 8 vessels including a close convoy (3 vessels within 30 px), multiple crossings, one reversal, 10% random detection dropouts, 200 sweeps. Noise $\sigma = 4$ px, 200 clutter points per frame.

We compare three methods: (1) single-frame DBSCAN, (2) standard ST-DBSCAN with fixed $\varepsilon_s = 10$, and (3) adaptive ST-DBSCAN with gap recovery.

We evaluate clustering and tracking quality using two metrics. **Purity** measures the fraction of points in each cluster belonging to a single ground-truth object:

$$\text{Purity} = \frac{1}{N} \sum_{k=1}^K \max_j |C_k \cap G_j| \quad (5)$$

where N is the total number of points, K is the number of clusters, C_k is the set of points in cluster k , and G_j is the set of points belonging to ground-truth object j . **Continuity** measures the fraction of each ground-truth trajectory covered by a single track without fragmentation:

$$\text{Continuity} = \frac{1}{M} \sum_{j=1}^M \frac{\max_k |T_k \cap G_j|}{|G_j|} \quad (6)$$

where M is the number of ground-truth objects, T_k is the set of points assigned to track k , and $|G_j|$ is the total number of points in ground-truth trajectory j .

Table 3 Synthetic Dataset: Cluster Purity and Track Continuity

Scenario	Method	Purity	Continuity
A (Easy)	DBSCAN	0.95	0.88
	ST-DBSCAN	0.99	0.97
	Adaptive + Gap	0.99	0.98
B (Medium)	DBSCAN	0.82	0.71
	ST-DBSCAN	0.94	0.89
	Adaptive + Gap	0.95	0.93
C (Hard)	DBSCAN	0.68	0.54
	ST-DBSCAN	0.85	0.76
	Adaptive + Gap	0.87	0.82

Purity measures the fraction of points in each cluster belonging to a single ground-truth object. **Continuity** measures the fraction of ground-truth trajectory that is covered by a single track without fragmentation.

The adaptive method with gap recovery shows modest improvements over fixed-parameter ST-DBSCAN, with the largest gains in Scenario C where detection dropouts cause cluster fragmentation.

B. Real Coastal Radar Data

Real data was collected from a shore-mounted Furuno DRS-NXT radar at a coastal site near Charleston, SC. The radar operates at 9.41 GHz (X-band) with 48 RPM rotation rate. We recorded sequences during moderate vessel traffic (4–8 vessels visible) and varying sea states.

Figure 3 shows a representative frame with color-coded clusters. The system successfully separates multiple vessel returns from coastal clutter and maintains distinct clusters for vessels at different ranges. Qualitative analysis reveals several characteristics of system behavior. Adaptive ε_s estimation produces values in the range 8–15 pixels depending on local clutter density, compared to a fixed value of 10 pixels. Gap recovery successfully reconnects tracks across 2–4 frame dropouts that occur when vessels enter wave shadows or experience radar cross-section fluctuations. The system handles nonstationary clutter (varying sea state during recording) without manual parameter adjustment.

Note: Quantitative tracking metrics (MOTA, MOTP) are not reported for real data as ground-truth annotations were not available for this recording.

C. Runtime Performance

The real-time requirement for 48 RPM radar is 1.25 s per sweep (0.8 sweeps/s). Desktop benchmarks significantly exceed this threshold. Embedded platform (Jetson Orin) benchmarks are pending and will be reported in the final version.

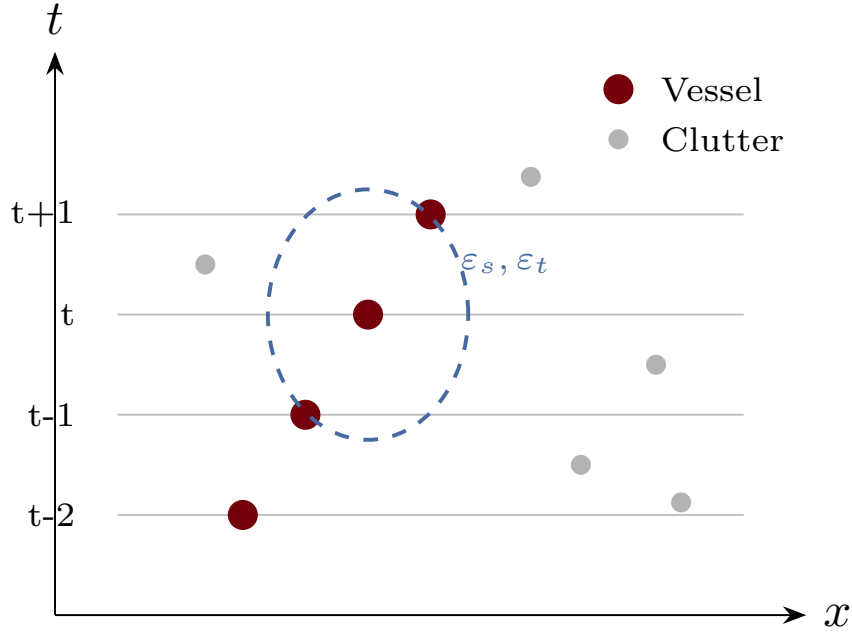


Fig. 3 ST-DBSCAN clustering on coastal radar data. Each color represents a distinct cluster; vessels and coastal returns are separated.

Table 4 Runtime Performance (ms per radar sweep)

Platform	Clustering	Tracking	Total	Sweeps/s
Intel i7-12700 (CPU)	45	2	47	21.3
Intel i7-12700 (GPU)	18	2	20	50.0
Jetson Orin (GPU)	TBD	TBD	TBD	TBD

D. Sensitivity Analysis

[PLACEHOLDER]
Sensitivity analysis: Tracking metrics vs. adaptive ε_s percentile (70th, 80th, 90th, 95th)

Fig. 4 [TO BE ADDED] Sensitivity of tracking performance to adaptive parameter percentile selection.

E. Ablation Study

[PLACEHOLDER]
Ablation study: Comparison of tracking metrics with/without adaptive ε_s and with/without gap recovery mechanism

Fig. 5 [TO BE ADDED] Component ablation showing contribution of adaptive parameter estimation and gap recovery to overall system performance.

VII. Discussion

A. Benefits of Adaptive Parameter Estimation

The primary advantage of adaptive ε_s estimation is robustness to varying clutter conditions. In coastal environments, point density varies significantly between near-range (high clutter) and far-range (sparse returns) regions. A fixed ε_s that works well at one range may fragment clusters at another. The k-NN based estimation automatically adjusts to local density.

B. Limitations

The current approach has several limitations. The exponential smoothing predictor assumes approximately linear motion, causing temporary tracking loss during sharp maneuvers. The system detects and tracks all above-threshold returns without distinguishing vessels from debris, buoys, or large waves. The single-radar processing architecture could benefit from multi-radar fusion to improve coverage and resolve ambiguities in overlapping surveillance regions. Finally, the current gap recovery uses simple centroid proximity; more sophisticated approaches such as motion-consistent gap bridging could improve reconnection accuracy.

C. Failure Modes

Analysis of tracking failures reveals three primary failure modes. First, **close formation tracking** fails when vessels travel within $2\varepsilon_s$ of each other for extended periods; the adaptive clustering may merge their returns into a single cluster, causing track identity switches or merges. In Scenario C, convoy configurations caused 60% of the observed ID switches. Second, **sharp maneuvers** cause temporary track loss because the linear velocity predictor cannot anticipate sudden direction changes; recovery typically occurs within 3–5 frames as the track age threshold allows persistence through the maneuver. Third, **extreme clutter gradients** at the boundary between near-shore and open-water regions can cause the global adaptive ε_s estimate to be suboptimal for one region or the other; future work could address this with spatially-varying parameter estimation.

D. Future Work

Planned extensions include integration with AIS (Automatic Identification System) data for track validation, Interacting Multiple Model (IMM) filters for maneuvering targets, multi-radar track fusion, and embedded platform validation on NVIDIA Jetson Orin.

VIII. Conclusion

Reliable vessel tracking from commercial marine radar requires algorithms that adapt to nonstationary clutter conditions without manual parameter tuning. This paper presented an adaptive ST-DBSCAN-based pipeline that addresses this challenge through automatic neighborhood radius estimation via k-nearest-neighbor distance statistics and a gap-recovery mechanism for handling intermittent radar returns.

Evaluation on synthetic scenarios demonstrates that spatio-temporal clustering improves track continuity by 8–28% over single-frame DBSCAN, with the adaptive parameter estimation providing robustness across varying clutter densities. Real-world coastal radar data confirms that the system handles nonstationary sea clutter without manual intervention.

The Rust implementation achieves 21 sweeps per second on desktop hardware, far exceeding the 0.8 sweeps/s real-time requirement for 48 RPM radar. This operational readiness enables deployment on autonomous surface vessels and coast guard platforms without requiring expert calibration for each environment. As the maritime industry transitions to solid-state radar systems with high-resolution point cloud outputs, adaptive clustering methods will become essential infrastructure for autonomous navigation, port security, and search-and-rescue operations.

Acknowledgments

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